

Domain and Range: All Three Notations - Practice

Algebra II | Unit 1 | Lesson 1 Practice | TEK A2.7.I

Objective: I can find domain and range from pairs, tables, and graphs, convert between all three notations, and tell continuous from discrete.**Section 1 — Recall**

- 1.
- $\{(-6, 2), (-1, 5), (3, 5), (8, -4)\}$

Domain: { _____ } Range: { _____ }

2.

x	-4	-2	0	3	6
f(x)	10	2	-2	4	16

Domain: { _____ } Range: { _____ }

3.

x	-5	-3	-1	2	4
f(x)	8	8	-6	3	11

Domain: { _____ } Range: { _____ }

4. Continuous or discrete?

- a. the number of pencils sold in a day _____
- b. the height of a plant over one growing season _____
- c. the cost of buying n notebooks at \$2 each _____

Section 2 — Apply

5. Convert "all real numbers less than or equal to -3" into inequality, set notation, and interval notation.

Inequality: _____

Set notation: $\{x \mid x \in \mathbb{R}, \text{_____}\}$

Interval notation: _____

6. Convert the interval
- $[2, 9)$
- into inequality notation and a verbal description.

Inequality: _____

Verbal: _____

7. Convert
- $4 < x \leq 10$
- into set notation and interval notation. Does infinity appear in the interval?

Set notation: $\{x \mid x \in \mathbb{R}, \text{_____}\}$

Interval notation: _____ Infinity appears? _____

8. Convert the interval
- $(-\infty, 0)$
- into inequality notation, set notation, and a verbal description.

Inequality: _____

Set notation: $\{x \mid x \in \mathbb{R}, \text{_____}\}$

Verbal: _____

Section 3 — Challenge

9. A graph shows a ray starting at the closed point $(-2, 5)$ and rising without bound as x increases, with no break anywhere. Give the domain in inequality notation and the range in interval notation.

Domain: _____ Range: _____

10. A graph shows a ray with an open circle at $(-1, -3)$ that extends left and upward forever as x decreases, with no break anywhere. Give the domain in inequality notation and the range in interval notation.

Domain: _____ Range: _____

11. Relation A is a solid horizontal ray starting at $(1, 4)$ and extending right forever (every point included). Relation B is the four discrete points $(1,4)$, $(2,4)$, $(3,4)$, $(4,4)$. They share the same range. Give each domain in the notation that fits, and explain in one sentence why one of them cannot use interval notation.

Domain A: _____ Domain B: { _____ }

Why not B: _____

Section 4 — Explain It

12. A classmate says: "Since $\{-2, 0, 4\}$ is a set of three numbers between -2 and 4 , I can also write its domain as the interval $(-2, 4)$." Explain what is wrong with this reasoning. Use the words *continuous* and *discrete* in your explanation.
